Momentum Riesz Defects and Whole-Interval Certificates for Concave-Cometric Neck Returns

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M687

October 9, 2026

Abstract

The assigned manuscript constructs a closed three-manifold with no conjugate points and a topological obstruction to every nonpositively curved metric. Its analytic mechanism is an exact coordinate index identity on a concave cometric neck, followed by an endpoint minimization that is uniform over returning geodesics. We retain that minimization and compute a more precise momentum Riesz norm. A positive scalar boundary solution supplies an exact defect in the source's upper bound, without division by the geodesic's vanishing interior normal velocity. We give a whole-interval residual and quadrature certificate for this defect, including boundary errors, derivative jumps and the quadratic cost of long intervals. A noncommuting concave cometric example tests the formulas, while negative controls show why boundary-free or finite-sample certificates are insufficient. The topological argument, smooth gluing and the uniform all-geodesic construction are kept separate from these local deductions. No larger universal construction parameter or new global existence theorem is asserted. Contribution assessment: C2/5, the contribution is a modest exact refinement and conditional certification of the source's momentum bound, not a replacement of its endpoint minimization or global geometry.

Contents

1 The source problem and its distinct obligations **2**

2 Physical neck, return and endpoint operator **3**
2.1 The cometric and its scales 3
2.2 Returning geodesics 3
2.3 The source's exact minimization 4

3 An exact momentum Riesz defect **4**

4 What the defect adds to a passage estimate **6**

5 Whole-interval certification, with its costs **6**
5.1 Negative controls 7

6 Exact calibration and a noncommuting neck **8**
6.1 Constant scalar potential 8
6.2 A physical finite-dimensional interface 9

7 Global interfaces that are not replaced **9**

8 Status, advantages and further work **9**

9 Reading and reproducibility boundaries **10**

1 The source problem and its distinct obligations

The source [1] asserts the existence of a closed connected orientable smooth three-manifold M carrying a metric without conjugate points, although M carries no metric of nonpositive sectional curvature. These are different quantifiers: the constructed metric must control every complete geodesic and every compactly supported variation; the curvature obstruction excludes every possible nonpositively curved metric on the same manifold.

The underlying graph manifold consists of two pieces $N_i = \Sigma_i \times S^1$, with Σ_i a once-punctured torus. Write (h_i, f_i) for its boundary and fiber classes. The gluing relations are

$$h_2 = h_1 + f_1, \qquad f_2 = h_1. \tag{1}$$

They have determinant -1 in the boundary bases. In a hypothetical geometric CAT(0) action, the boundary $\mathbb{Z}^2$ has a Euclidean translation plane. The centralizer translation character vanishes on the commutator h_i; thus h_i and f_i have orthogonal translation vectors. If u, v represent h_1, f_1, the two pieces would give $u \perp v$ and $u+v \perp u$, hence $|u|^2 = 0$. This contradicts the rank-two translation lattice. The source supplies the semisimplicity argument and invokes the flat torus/minset machinery. We do not replace those group-action theorems by a finite matrix computation.

Scientific architecture. The introduction and topology module specify the manifold and the obstruction. The cometric module constructs the neck, fixes its two physical end frames, proves concavity and derivative control, and matches open collar germs. The variations module derives the exact intrinsic index identity, classifies neck passages and selects passage-dependent tangential gauges. The turning module proves an exact endpoint-operator formula and a uniform small-parameter estimate. The global module sums the original index integrand over possibly infinitely many passages and turns nonnegative index into absence of conjugate points. The punctured-torus appendix gives an explicit hyperbolic cusp model. The neck figure, driver, bibliography and README were also read. This report's new argument is within the turning module; it is not a certification of every global dependency.

Original Ram comparison. The original boundary-value report [2], in the passages specified in Section 9, distinguishes a differential recurrence from an invertible boundary operator, includes physical flux matching, and requires a genuine inverse bound for residual-to-error transfer. Its Dirichlet resonance example is an especially relevant warning. The original physical-metric chapter [3] uses the actual synthesis Gram form and retains the discarded residual component. Here the actual forms are Q on vector velocities and the Dirichlet energy in (10); no Euclidean coefficient norm is substituted for either one.

2 Physical neck, return and endpoint operator

2.1 The cometric and its scales

In one of the source's fixed end frames the neck metric is

$$g = dt^2 + dx^T Q(t)\, dx, \qquad Q = P^{-1}, \qquad x \in \mathbb{R}^2/\Lambda. \tag{2}$$

For orientation, the source constructs

$$a(u) = \begin{cases} e^{-1/(2/3-u)}, & u < 2/3, \\ 0, & u \geq 2/3, \end{cases} \quad A(u) = \int_0^u a, \quad \epsilon = \delta^2,$$

$$P(t) = (1 + \delta A(t)) e_1 e_1^T + (1 + \delta A(1-t))(e_2 + \epsilon\chi(t) e_1)(e_2 + \epsilon\chi(t) e_1)^T.$$

Here χ is a fixed smooth nondecreasing transition, zero before $5/12$ and one after $7/12$. The periods satisfy $B_1 = \epsilon B_2$. The right end uses the frame $(e_2 + \epsilon e_1, e_1)$, and its cometric transforms by congruence. These changes of frame need not be Euclidean isometries.

For sufficiently small fixed δ, the source proves in both end frames uniform positive bounds for P, Q, $-P'' \succeq 0$, and

$$|P'p|^2 \leq C_0 \delta\, p^T(-P'')p. \tag{3}$$

No strict lower bound on $-P''$ is required throughout the neck. At an inward end coordinate $t = 0$,

$$P(0)e = e, \quad P'(0) = \kappa e e^T, \quad e = e_1, \quad \kappa = \delta a(0) > 0. \tag{4}$$

The construction first chooses δ for the analytic inequalities, then chooses the periods small enough to fit hyperbolic cusp collars. Exact agreement on open germs, rather than a finite jet match, gives the smooth glued metric.

2.2 Returning geodesics

For a unit-speed geodesic $\gamma(s) = (t(s), x(s))$ in the neck, the covector $p = Qx'$ is constant. Set $F(t) = p^T P(t) p$ and $v = t'$. The geodesic equations give

$$v^2 = 1 - F(t), \qquad v' = -\tfrac{1}{2}F'(t), \qquad v'' = -\tfrac{1}{2}F''(t)v. \tag{5}$$

A prime on F denotes the t derivative, and a prime on v denotes the s derivative. For a nonconstant return to the same boundary, choose the inward coordinate and translate time so that

$$t(0) = t(L) = 0, \quad t(L/2) = t_* > 0, \quad v(0) = -v(L) = v_0 > 0.$$

The source proves that there is exactly one turning point, $t(L-s) = t(s)$, $v(L-s) = -v(s)$, and

$$0 < F'(t) \leq \kappa p_1^2 \ (0 \leq t \leq t_*), \quad p_1 \neq 0, \quad v_0^2 \leq \kappa p_1^2. \tag{6}$$

In particular

$$\mathcal{V}(s) = -\tfrac{1}{2}F''(t(s)) \geq 0 \tag{7}$$

is smooth and symmetric about $L/2$. Nothing below requires division by $v(s)$ at the turning point.

2.3 The source's exact minimization

For a variation $V = (w, z)$ put $\xi = z' - P'p\,w$. The coordinate energy is

$$\mathcal{E}(V) = \int_0^L \left(w'^2 + \xi^T Q \xi + \mathcal{V} w^2\right) ds. \tag{8}$$

The intrinsic index differs by its endpoint Christoffel term. The source uses tangential gauge invariance of $I^\perp(V) = I(V) - \int \langle V, T\rangle'^2$ to impose $w(0) = w(L) = 0$ and an affine tangential component. For $X = z(L) - z(0)$ it obtains

$$I^\perp(V) \geq \mathcal{E}(V) + \mu (e^T X)^2 - \frac{(p^T X)^2}{L}, \qquad \mu = \frac{\kappa v_0}{4}. \tag{9}$$

The endpoint sign comes from the inward normal velocities $v_0, -v_0$. Different passage gauges are not asserted to fit together globally.

Use the real Hilbert space

$$\mathcal{D} = H_0^1(0, L), \qquad \langle w, z\rangle_{\mathcal{D}} = \int_0^L (w'z' + \mathcal{V} w z)\, ds, \quad \Phi_q(w) = \int_0^L q^T P' p\, w\, ds. \tag{10}$$

The source's exact endpoint operator is

$$S(q, r) = \int_0^L q^T P r\, ds + \langle \Phi_q, \Phi_r\rangle_{\mathcal{D}^*}. \tag{11}$$

Indeed $X = \int (\xi + P'p\,w)$ and Riesz duality in $L^2(Q) \oplus \mathcal{D}$ gives this Gram form. Completing the single endpoint penalty then gives, exactly,

$$(p^T X)^2 \leq m\,(\mathcal{E} + \mu (e^T X)^2), \qquad m = S(p, p) - \frac{S(p, e)^2}{S(e, e) + \mu^{-1}}. \tag{12}$$

The minimizer is $S(p, e)/(S(e, e) + \mu^{-1})$. Both (11) and (12) are already source results. Our refinement computes an additional scalar defect inside $S(p, p)$.

3 An exact momentum Riesz defect

Theorem 3.1 (Positive boundary solution and exact defect)**.** *Assume the actual returning-geodesic hypotheses (5)–(7). Let u solve*

$$-u'' + \mathcal{V} u = 0, \qquad u(0) = u(L) = 1. \tag{13}$$

Then u is positive, symmetric and convex, with $u \leq 1$. Define

$$\Gamma = -\frac{1}{v_0} \int_0^L v u'\, ds, \qquad u_{\min} = u(L/2). \tag{14}$$

The Riesz representative of Φ_p in $\mathcal{D}$ is

$$\psi_p(s) = (s - L/2) v(s) + \frac{L v_0}{2} u(s). \tag{15}$$

Consequently

$$\|\Phi_p\|_{\mathcal{D}^*}^2 = \int_0^L v^2\,ds + Lv_0^2(1-\Gamma), \tag{16}$$

$$S(p,p) = L + Lv_0^2(1-\Gamma), \tag{17}$$

$$0 \le \Gamma \le 1 - u_{\min}^2 < 1. \tag{18}$$

Moreover $\Gamma = 0$ *exactly when* $\mathcal{V}$ *is identically zero.*

Proof. The bilinear form in (10) is coercive on H_0^1 by Poincare's inequality. Lifting the boundary value 1 gives a unique weak solution of (13); ordinary one-dimensional regularity makes it smooth. The maximum principle gives $0 < u \le 1$. Uniqueness gives symmetry, and $u'' = \mathcal{V}u \ge 0$ gives convexity. Thus $u' \le 0$ on the first half and $u' \ge 0$ on the second.

Write $\mathcal{L} = -d^2/ds^2 + \mathcal{V}$. Equation (5) says $\mathcal{L}v = 0$, so $\mathcal{L}((s-L/2)v) = -2v' = F'(t(s))$. The expression in (15) vanishes at both endpoints and $\mathcal{L}u = 0$. Testing against any $w \in H_0^1$ therefore proves $\langle \psi_p, w\rangle_{\mathcal{D}} = \Phi_p(w)$. Integration by parts, using these zero endpoints, yields

$$\|\Phi_p\|_{\mathcal{D}^*}^2 = \int F'\psi_p = 2\int v\psi_p' = \int v^2 + Lv_0^2 + Lv_0 \int vu'.$$

Here the boundary contribution in $\int (s-L/2)vv'$ is $Lv_0^2/2$. Since $\int F = L - \int v^2$, (16) and (17) follow.

On the first half $v > 0$ and $u' \le 0$, so $\Gamma \ge 0$. The function $u - v/v_0$ solves $\mathcal{L}(u - v/v_0) = 0$ there and has boundary values $0, u_{\min} > 0$. The maximum principle gives $v/v_0 \le u$. Symmetry then shows

$$\Gamma = \frac{2}{v_0}\int_0^{L/2} v(-u')\,ds \le 2\int_0^{L/2} u(-u')\,ds = 1 - u_{\min}^2.$$

If $\Gamma = 0$, continuity and $v > 0$ on the open first half force $u' = 0$ there. Symmetry gives $u \equiv 1$, so $\mathcal{V} = u''/u \equiv 0$. The converse follows directly. □

The source first bounds $\|\Phi_p\|_{\mathcal{D}^*}^2 \le 4\int v^2$, then uses convexity of v on the first half to obtain $\int v^2 \le Lv_0^2/3$. This gives $S(p,p) \le L(1+v_0^2)$. Formula (17) retains the nonnegative amount $Lv_0^2\Gamma$ that this bound discards. It is an identity for the actual whole-interval operator, not an assertion that a finite Gram matrix determines it.

Corollary 3.2 (A derivative-free normalized formula)**.** *The measure*

$$d\nu(s) = \frac{F'(t(s))}{4v_0}\,ds \tag{19}$$

is a probability measure on $[0, L]$*, and*

$$\Gamma = 2 - 2\int u\,d\nu. \tag{20}$$

Proof. Integrating $F' = -2v'$ gives $\int F' = 4v_0$ and positivity follows from (6). Also $\int vu' = [vu]_0^L - \int v'u = -2v_0 + \frac{1}{2}\int F'u$. Substitution gives (20). □

The normalized formula avoids differentiating an approximate u in the final measurement. It does not make the law ν known from samples: its normalization and its dependence on the actual trajectory still require control, particularly as v_0 tends to zero.

4 What the defect adds to a passage estimate

For returns reaching $t_* > 1/4$, the source proves

$$4v_0 \le \beta\kappa p_1^2 L, \quad \beta = \frac{1+e^{-9/26}}{2} < 1, \quad S(e,e) \le C_e L, \quad |S(p,e) - Lp_1| \le C_m \delta L |p_1|.$$

The last estimate has $|p_1|$, not $|p|$. Replacing it by an unscaled error would lose the shallow-return limit. The constants can be taken as $C_e = 3C_0$ and $C_m = a(0) + \sqrt{2C_0 a(0)} + 2\sqrt{2C_0 a(0)/3}$ in the source.

Proposition 4.1 (Conditional margin with a certified defect)**.** *Retain all of these source inequalities,* (3)*, and* $C_m\delta \le 1/2$*. Suppose a whole-interval certificate gives* $0 \le \gamma_* \le \Gamma$*. Put*

$$G_\delta = \frac{(1-C_m\delta)^2}{\beta + C_e a(0)\delta}, \qquad \sigma = G_\delta + \gamma_* - 1.$$

If $\sigma \ge 0$*, the actual gauged variation satisfies*

$$I^\perp(V) \ge v_0^2 \sigma \left(\mathcal{E}(V) + \mu (e^T X)^2\right). \tag{21}$$

Proof. Writing $r = v_0^2/p_1^2$, the source inequalities give $r \le a(0)\delta$ and $L\mu \ge r/\beta$. Hence

$$\frac{S(p,e)^2}{S(e,e) + \mu^{-1}} \ge \frac{L^2 p_1^2 (1-C_m\delta)^2}{C_e L + L\beta/r} \ge L v_0^2 G_\delta.$$

Theorem 3.1 and (12) imply $m \le L[1 - v_0^2\sigma]$. Substitute this in (9). □

The source chooses a single δ so that $G_\delta \ge 1$ for every eligible return. That proof already handles arbitrarily small p_1 and arbitrarily large L. Proposition 4.1 can improve the account of an individual return, or of a family with a separately proved uniform lower bound on Γ. We have not proved such a family bound for a larger δ. Even when $\sigma > 0$, its physical coercivity coefficient $v_0^2\sigma$ may tend to zero. Returns contained in the end collar use the source's nonpositive-curvature argument instead.

5 Whole-interval certification, with its costs

Theorem 5.1 (Boundary, panel and jump residuals)**.** *Let* $\mathcal{V} \ge 0$ *be continuous on* $[0, L]$ *and let* u *solve* (13)*. Let* $\widehat{u}$ *be continuous and piecewise* C^2 *on a finite partition, with one-sided derivatives at its internal nodes* s_j*. Suppose, on the* whole *open panels,*

$$|\widehat{u}(0) - 1|, \ |\widehat{u}(L) - 1| \le \varepsilon_b, \qquad |-\widehat{u}'' + \mathcal{V}\widehat{u}| \le \varepsilon_r.$$

Put $J_j = \widehat{u}'(s_j+) - \widehat{u}'(s_j-)$ *and*

$$\eta = \varepsilon_b + \frac{L^2}{8}\varepsilon_r + \frac{L}{4}\sum_j |J_j|. \tag{22}$$

Then $\|u - \widehat{u}\|_\infty \le \eta$*. If* $\widehat{\nu}$ *is a supplied probability approximation to the actual law* (19) *and*

$$\left| \int \widehat{u}\, d\nu - \int \widehat{u}\, d\widehat{\nu} \right| \le \varepsilon_\nu,$$

then, with $\widehat{\Gamma} = 2 - 2\int \widehat{u}\, d\widehat{\nu}$*,*

$$|\Gamma - \widehat{\Gamma}| \le 2\eta + 2\varepsilon_\nu, \qquad \gamma_* = \max\{0, \widehat{\Gamma} - 2\eta - 2\varepsilon_\nu\} \le \Gamma. \tag{23}$$

Proof. The Dirichlet Green kernel of $-d^2/ds^2 + \mathcal{V}$ is positive and bounded by that of $-d^2/ds^2$:

$$0 \le G_{\mathcal{V}}(s,t) \le G_0(s,t) = \frac{\min(s,t)(L - \max(s,t))}{L}.$$

For completeness, inverse positivity follows from the weak maximum principle. If $f \ge 0$ and $z_0 = (-d^2/ds^2)^{-1} f$, then $\mathcal{L} z_0 = f + \mathcal{V} z_0 \ge f$, proving inverse domination; approximation of point sources gives the kernel statement. The homogeneous solution with endpoint errors has absolute value at most ε_b. Distributionally, the residual of $\widehat{u}$ is its panel residual minus $\sum J_j \delta_{s_j}$. Green representation and

$$\int_0^L G_0(s,t)\, dt = \tfrac{1}{2} s(L-s) \le L^2/8, \qquad G_0(s,t) \le L/4$$

give (22). Probability normalization gives $|\int (u - \widehat{u}) d\nu| \le \eta$, after which (20) proves (23). □

Coefficient and quadrature interfaces. If only an approximation $\widehat{\mathcal{V}}$ is used in computation, a valid whole-interval estimate is

$$\varepsilon_r \le \| - \widehat{u}'' + \widehat{\mathcal{V}} \widehat{u} \|_\infty + \|\mathcal{V} - \widehat{\mathcal{V}}\|_\infty \|\widehat{u}\|_\infty.$$

The positivity assumption concerns the actual $\mathcal{V}$, not a few matrix samples. If a partition has exact ν masses and a quadrature point in each cell of diameter at most h, then $\varepsilon_\nu \le h \|\widehat{u}'\|_\infty$. If the discrete masses have ℓ^1 error ε_w, an additional $\varepsilon_w \|\widehat{u}\|_\infty$ suffices. These are sufficient, not optimal, bounds and require the original trajectory law.

On the unit interval $x = s/L$, the residual is $-\widehat{U}_{xx} + L^2 \mathcal{V}(Lx) \widehat{U}$ and the derivative jump is $L J_j$. Thus rescaling exposes rather than removes the long-interval cost. A verified positive lower bound on $\mathcal{V}$ could reduce this cost, but the source permits degenerate $-P''$ and supplies no such uniform lower bound over all returns.

5.1 Negative controls

The following failures are within the very same scalar boundary problem. For $\mathcal{V} = 0$ the exact solution is $u = 1$.

1. Taking $\widehat{u} = 0$ gives zero differential residual and error 1. Boundary conditions cannot be omitted.

2. Taking $\widehat{u} = 1 + \varepsilon s(L-s)/2$ gives residual ε and error $\varepsilon L^2/8$. The quadratic cost in (22) is sharp in this admissible class.

3. A continuous tent added to 1, zero at both endpoints, has zero second derivative on each open panel. Its nonzero error is detected only by the derivative jump; panelwise residuals alone miss it.

4. A smooth bump supported between finitely many sample points can leave all sampled jets zero. Sampled residuals are not a bound on the whole-panel residual. Valid Taylor or interval remainder enclosures are additional hypotheses.

The last point is precisely the distinction between an exact finite-class jet calculation and boundary-operator control in [2]. Neither finite data nor a chosen coefficient basis supplies an infinite-dimensional inverse theorem.

6 Exact calibration and a noncommuting neck

6.1 Constant scalar potential

For $\mathcal{V} = \omega^2 > 0$, let $z = \omega L/2$ and choose $v_0 > 0$. The solutions are

$$v(s) = v_0 \frac{\sinh(\omega(L/2 - s))}{\sinh z}, \qquad u(s) = \frac{\cosh(\omega(s - L/2))}{\cosh z}.$$

Direct integration in (14) gives

$$\Gamma = 1 - \frac{2z}{\sinh(2z)}, \qquad S(p,p) = L + Lv_0^2 \frac{2z}{\sinh(2z)}. \tag{24}$$

For instance, integrate $\sinh^2(\omega(s - L/2))$ using $2\sinh^2 x = \cosh(2x) - 1$. The improvement is strict, but tends to zero as $z \to 0$: $\Gamma = 2z^2/3 + O(z^4)$. This is another reason that positivity alone is not a uniform quantitative gain. For large z, $\Gamma \to 1$. The limiting case $\omega = 0$ gives $u = 1$, linear v, and $\Gamma = 0$.

Example 6.1 (An actual concave matrix model)**.** On $0 \le t \le 1$ set

$$F(t) = \frac{9}{25} + 2t - t^2, \qquad p = (3/5, 0)^T, \qquad P(t) = \begin{pmatrix} 25F(t)/9 & t^2/4 \\ t^2/4 & 1 - t^2/4 \end{pmatrix}. \tag{25}$$

This is a positive cometric, since $P_{11} \ge 1$, $P_{22} \ge 3/4$ and $|P_{12}| \le 1/4$, giving $\det P \ge 11/16$. Also

$$-P'' = \begin{pmatrix} 50/9 & -1/2 \\ -1/2 & 1/2 \end{pmatrix} \succ 0, \qquad \det(-P'') = 91/36, \qquad P(0) = I, \quad P'(0) = \operatorname{diag}(50/9, 0).$$

This is not a simultaneously diagonal matrix family. For example $[P'(0), P(1)]_{12} = 25/18 \neq 0$. Let $z = \log 3$, $L = 2z$, and

$$t(s) = 1 - \frac{\cosh(s - z)}{\cosh z}, \qquad v(s) = -\frac{\sinh(s - z)}{\cosh z}.$$

Then $v_0 = 4/5$, $t_* = 2/5$, $p^T P p = F$, $v^2 = 1 - F(t)$ and $v' = -F'(t)/2$. Defining $x'(s) = P(t(s))p$ yields an actual unit-speed neck geodesic: its conserved momentum and normal equation are exactly the Euler–Lagrange equations of (2). The scalar potential is $\mathcal{V} = 1$ and

$$\Gamma = 1 - \frac{9}{20}\log 3, \qquad S(p,p) = 2\log 3 + \frac{72}{125}(\log 3)^2. \tag{26}$$

Here $2z/\sinh(2z) = 9\log 3/20$. Thus the exact excess over L is less than half of the source bound Lv_0^2.

The last comparison is rigorous without decimal rounding: $\log 3 < 10/9$, since the first five terms of $e^{10/9}$ already exceed 3. Consequently $9\log 3/20 < 1/2$. Example 6.1 is a local concave-cometric model, not the source's small-δ cutoff/shear family. It provides neither its open-germ gluing nor its topological manifold. Its purpose is to calibrate the physical identities, not to assert a new global metric.

6.2 A physical finite-dimensional interface

Suppose a finite trial space for (13) is synthesized by $Jc = \sum c_j \phi_j$ with $\phi_j \in H_0^1$. The Gram matrix is

$$G_{ij} = \int (\phi_i' \phi_j' + \mathcal{V} \phi_i \phi_j).$$

For independent trial functions it is positive definite. It represents the physical $\mathcal{D}$ norm, not the unweighted coefficient norm. A Ritz solution enforces only projected residual orthogonality. Theorem 5.1 requires the remaining differential and interface residuals on the entire interval. This is the original physical-metric principle of [3], here applied to a boundary solution with nonzero trace. One must first lift the trace 1 before using a homogeneous trial space. Changing basis does not remove the complement or prove a uniform inverse bound as $L \to \infty$.

7 Global interfaces that are not replaced

The preceding results concern one returning passage. Several correct source mechanisms must remain intact.

First, the intrinsic coordinate identity includes its endpoint term. The nonnegative expression (8) by itself is not the index. The gauge-invariant normal index, with its exact signs, is what permits (9). The penalty and its Schur complement are not optional numerical stabilizers.

Second, monotone passages can have infinite length and approach a constant-height geodesic. The source uses a tangential gauge only on the support of the variation; it does not assert a global uniform bound for $1/v$. Constant-height passages and returns entirely in an end collar are treated separately. Theorem 3.1 does not replace those cases.

Third, the source's countable-passage summation uses the *original* normal-index integrand. For a compactly supported smooth field on a complete geodesic this integrand is continuous and integrable. If J_i are the disjoint open neck passages, $\sum_i |I_{J_i}^{\perp}| \le \int |i^{\perp}|$. The complement, including possible positive-measure interface times, has nonnegative integrand by the curvature sign. This avoids any assumption of finitely many crossings. It would be invalid simply to glue the individually selected passage gauges and differentiate the resulting discontinuous field.

Finally, nonnegative index on every complete geodesic implies absence of conjugate points by a global test-field argument. A Jacobi field with two zeros can be extended by zero as an H^1 field. Its index is zero, and positivity of its perturbation by both signs of a test field forces its bilinear index pairing to vanish. Integration by parts then forces the nonzero endpoint derivative to vanish, a contradiction. Smooth density and parallel-frame approximation are needed to admit this H^1 test. A finite family of sampled geodesics cannot supply the premise of this argument.

8 Status, advantages and further work

Public status checked October 9, 2026. The assigned source is the September 24, 2026 manuscript in the OpenAI mathematics repository. Captured-tree byte identity and fresh path-history metadata are provenance, not mathematical acceptance. Leeb's primary abstract [4] describes closed graph-manifold obstructions to nonpositive curvature; it does not construct the no-conjugate-points metric analyzed here. Ivanov–Kapovitch [5] prove restrictions on centralizers and, in their Theorem 7.1, identify the nonpositive-curvature condition with existence of a no-focal-points metric in dimension three. Their Questions 8.1–8.2 distinguish the no-conjugate-points problem. We inspected these statements, not the complete external proof. The current arXiv record for the Burns–Matveev survey [6] is version 3, September 3, 2019, with a 2021 journal reference; its abstract

and version metadata were refreshed, not its entire proof content. None of these checks independently validates the assigned 2026 construction.

What is gained. The exact positive-solution identity identifies the amount omitted by a particular source estimate. Its normalized measurement avoids differentiating an approximate solution in the final defect, and the residual certificate lists actual boundary, whole-panel and interface requirements. The noncommuting example shows that scalar momentum analysis need not diagonalize the full cometric. The original Ram boundary and physical-form principles clarify precisely what a finite implementation would still have to certify.

Costs and limitations. Classical one-dimensional Green comparison, maximum principles and Riesz duality are credited as such. A normalized law is not free information about a shallow return. The factor L^2 in the residual budget can make long passages expensive. No performance measurement, optimality claim, uniform extra δ range, or new all-geodesic theorem is established. We found no source correction required in the scoped deductions; that is not an independent certification of all geometric or CAT(0) inputs.

Feasible extensions. A useful next step would be a whole-family lower bound on Γ expressed in terms of the time spent where $\mathcal{V}$ is positive, with explicit control as $p_1 \to 0$. Another would be a Green majorant adapted to the actual degenerate potential, improving the $L^2/8$ worst-case residual cost. Such results must be proved on the entire passage and combined with the signed $|p_1|$ cross estimate. They could then be inserted into the source's uniform proof. Until that work is done, optimizing a finite sample of the Schur margin is only a diagnostic.

9 Reading and reproducibility boundaries

Fresh pre-read hashes were persisted before literal displays. All 11 public text files were read, comprising 1,716 lines: nine TeX files, the bibliography and README. Their unread complements are empty. Full public reading does not mean full reading of the papers cited by that manuscript.

The original private TR212 has 1,053 lines. The fresh proof reads are 178–305, 444–526, 558–630 and 759–808, totaling 334 lines; the complementary 719 lines are explicitly unread in this task. The original volume 21 physical-metric chapter was read in full, lines 1–210. Thus the selected original-private proof scope is 544 lines, not the entire confidential corpus. The included external figure and cited references were not implicitly counted as reads. The two original documents remain local and were not sent to any public search service.

The owned checks supplement the written proofs. Exact algebra targets the boundary lift, concavity and noncommutativity of the matrix model, its conserved-energy identity, Green constants and negative controls. High-precision quadrature of the constant-potential model is separately classified as numerical, not as an interval certificate or theorem. The final run passed 41 exact/symbolic cases with 238 assertions and 24 numerical cases with 192 assertions at 70-digit working precision. These are finite model and operator-identity checks, not a proof by sampling. The script hash and execution receipt are recorded with the report. Neither parent checks nor foreign checkers are included in those counts.

References

[1] OpenAI mathematics project, *A Three-Manifold Without Conjugate Points and Without a Nonpositively Curved Metric*, September 24, 2026. Pinned local source, public manuscript directory. Scientific modules read as specified above.

[2] M. Subbarao, original confidential *TR212: Boundary-Value Ram Operators*, local `TR212-boundary-value-ram-operators.tex`. Proof passages 178–305, 444–526, 558–630, 759–808, freshly read; no whole-document proof audit claimed.

[3] M. Subbarao, original confidential *Ram Calculus Books*, volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Physical synthesis Gram forms, residual complements and domain conditions.

[4] B. Leeb, *3-manifolds with(out) metrics of nonpositive curvature*, Invent. Math. 122 (1995), 277–289. arXiv:dg-ga/9410002v1. Primary abstract and metadata refreshed; not a full external proof read.

[5] S. Ivanov and V. Kapovitch, *Manifolds without conjugate points and their fundamental groups*, J. Differential Geom. 96 (2014), 223–240. arXiv:1205.4455v1; author-hosted text. Selected theorem and question statements read.

[6] K. Burns and V. S. Matveev, *Open problems and questions about geodesics*, Ergodic Theory Dynam. Systems 41 (2021), 641–684. arXiv:1308.5417v3. Primary abstract and version record refreshed.